

MainCrafts Technology

Build & Evaluate a Linear Regression Model California Housing Price Predictor Artificial Intelligence & Machine Learning – Task 1

Organization: MainCrafts Technology

Author: Somya Jain

BUILD & EVALUATE A LINEAR REGRESSION MODEL *California Housing Price Predictor*

1. Objective

The objective of this project is to build and evaluate a Linear Regression model for predicting house values using the California Housing dataset.

The project demonstrates the complete machine learning workflow, including data loading, exploratory data analysis, preprocessing, train-test splitting, model training, prediction, evaluation, and visualization.

The model performance is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

2. Dataset

The California Housing dataset provided by scikit-learn was used for this project.

The dataset contains **20,640 samples and 8 input features** related to housing districts in California.

The input features are:

- MedInc — Median income
- HouseAge — Median house age
- AveRooms — Average number of rooms
- AveBedrms — Average number of bedrooms
- Population — Population of the district
- AveOccup — Average household occupancy
- Latitude — Geographic latitude
- Longitude — Geographic longitude

The target variable is **MedHouseVal**, which represents the median house value.

3. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the structure and characteristics of the dataset.

Missing Value Analysis

The dataset was checked for missing values using the `isnull().sum()` method. The analysis showed that the dataset did not contain missing values in the input features.

Target Variable Distribution

A histogram was plotted to understand the distribution of the target variable, `MedHouseVal`. The visualization helped in understanding the range and frequency of house values.

Correlation Analysis

A correlation matrix and heatmap were created to study relationships between the input features and the target variable.

The correlation analysis showed that **MedInc** has a relatively strong positive relationship with the median house value compared with several other features.

Figure 1: Correlation Matrix Heatmap

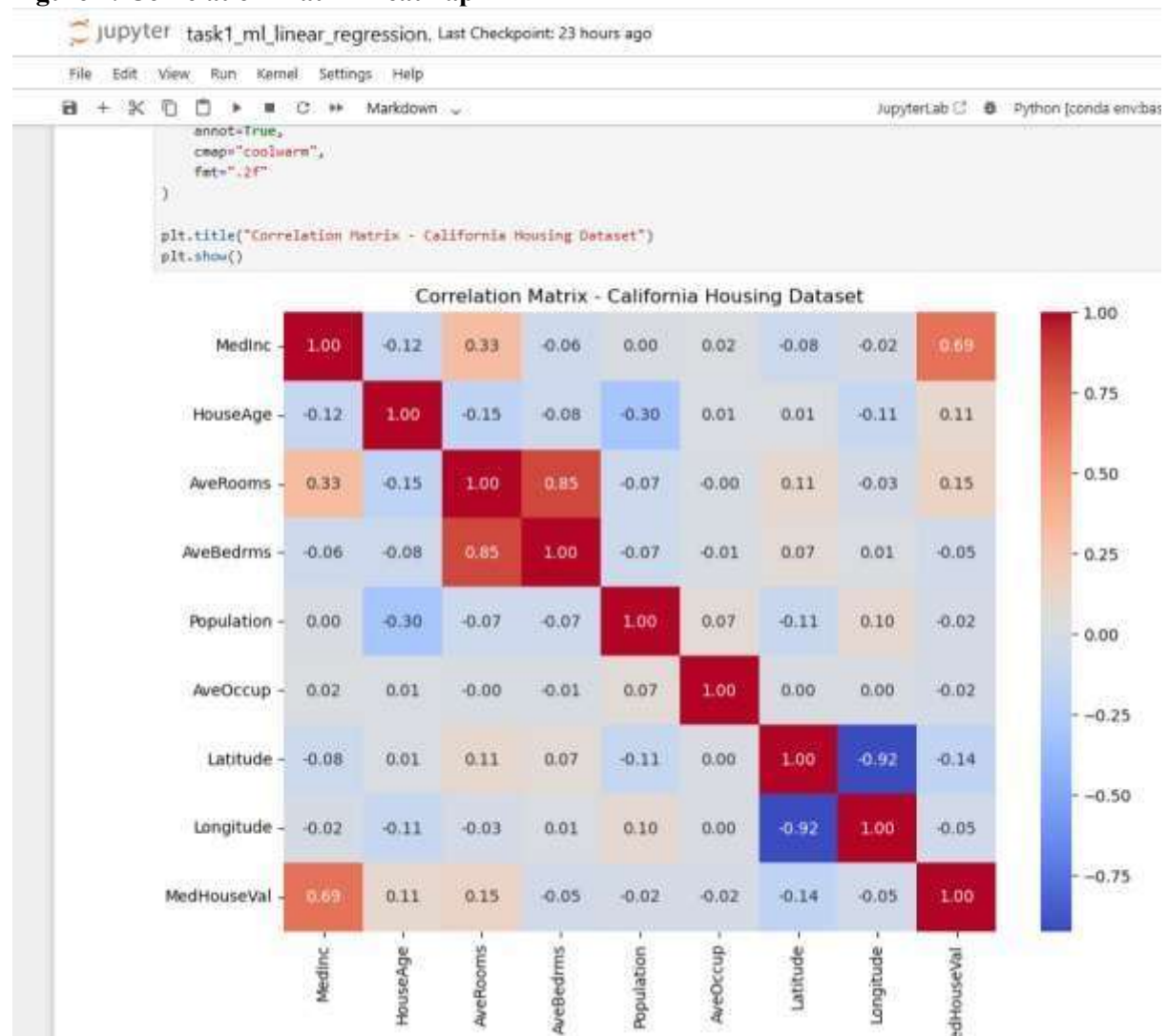
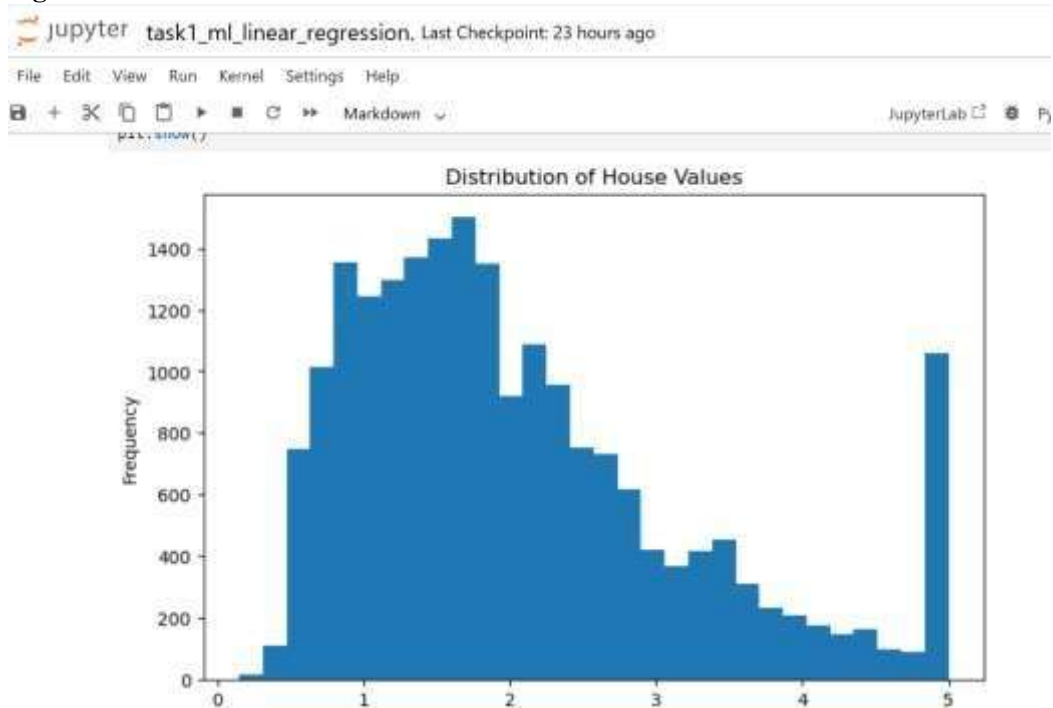


Figure 2: Distribution of House Values



4. Model Development

The dataset was divided into training and testing sets using an **80:20 split**.

- Training data: 80%
- Testing data: 20%
- Random state: 42

A Linear Regression model from scikit-learn was then trained using the training data.

The model learns the relationship between the eight input features and the target variable, MedHouseVal.

After training, predictions were generated on the test dataset.

5. Model Evaluation

The trained model was evaluated using three standard regression metrics.

Metric	Result	MAE	[ENTER YOUR MAE]	RMSE	[ENTER YOUR RMSE]	R ² Score	[ENTER YOUR R ²]	Interpretation
--------	--------	-----	------------------	------	-------------------	----------------------	------------------------------	----------------

Mean Absolute Error (MAE):

MAE represents the average absolute difference between the actual and predicted values. Lower MAE indicates smaller prediction errors.

Root Mean Squared Error (RMSE):

RMSE measures the square root of the average squared prediction errors. It gives more weight to larger errors. Lower RMSE indicates better performance.

R² Score:

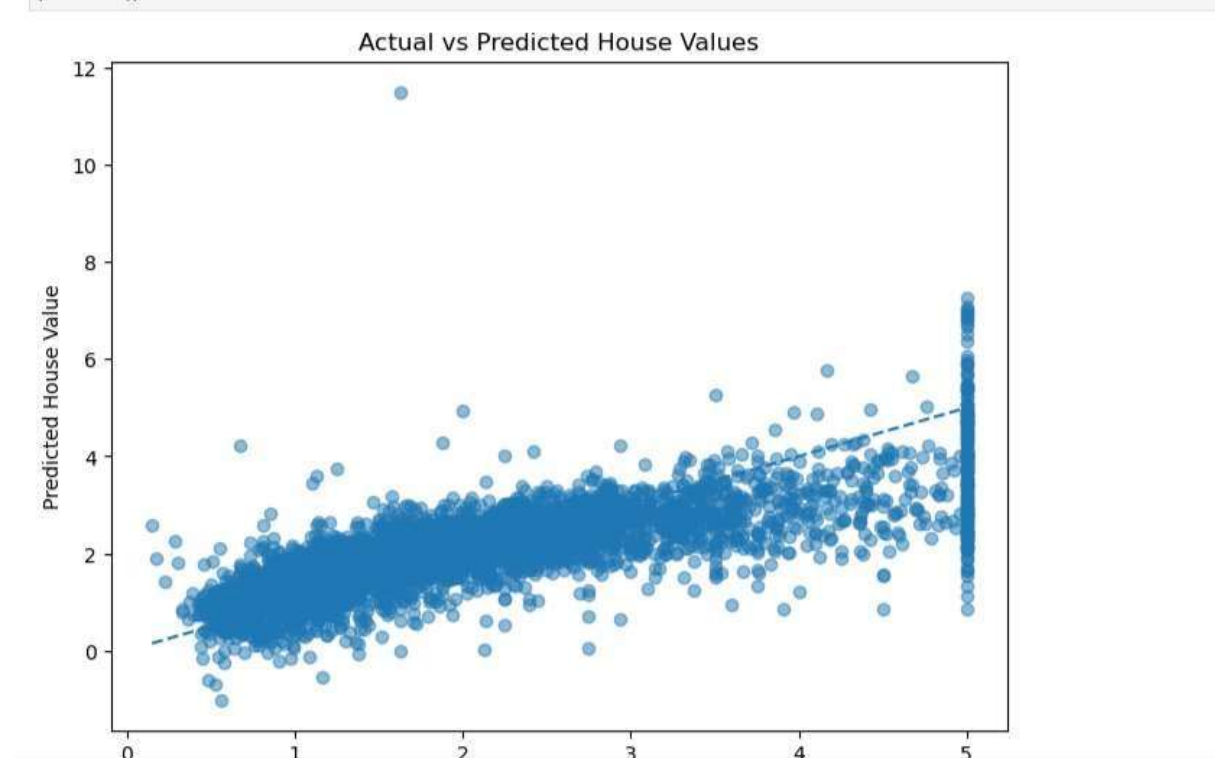
R² indicates how much of the variation in the target variable is explained by the model. A higher R² generally indicates a better fit to the data.

6. Prediction Visualization

An Actual vs Predicted plot was created to compare the model's predictions with the actual house values.

Points closer to the diagonal reference line indicate predictions that are closer to the actual values.

Figure 3: Actual vs Predicted House Values



A residual plot was also created to analyze the prediction errors.

Figure 4: Residual Plot

10. Residual Analysis

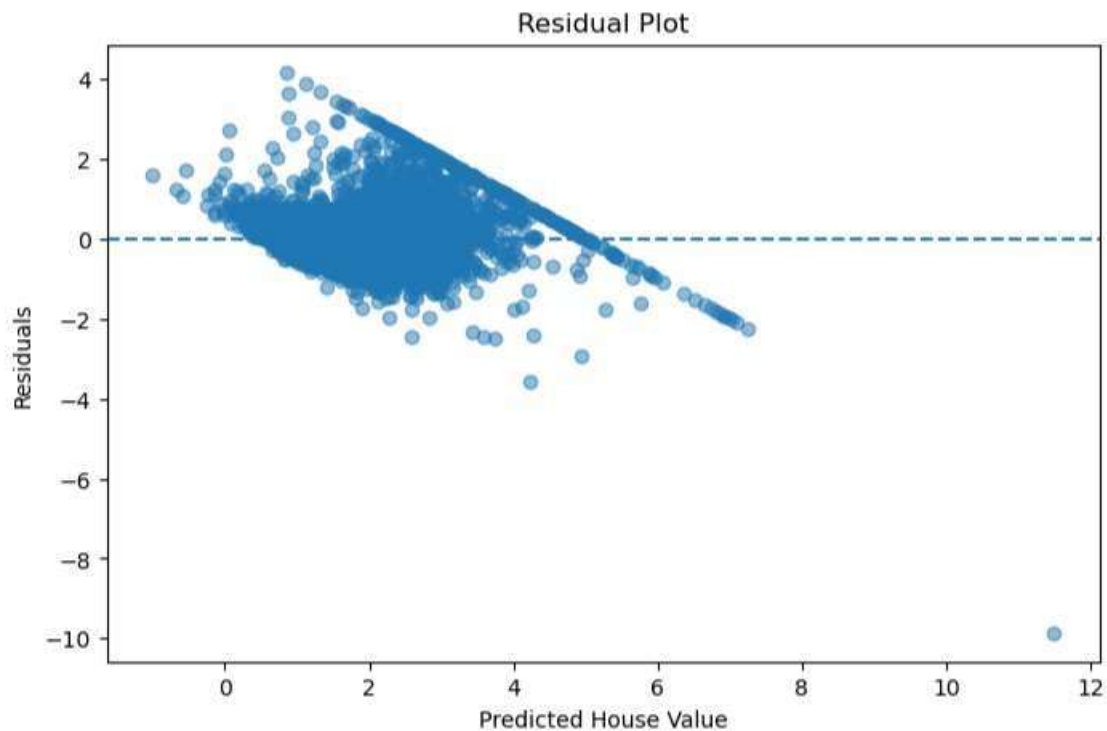
```
plt.figure(figsize=(8, 5))

plt.scatter(y_pred, residuals, alpha=0.5)

plt.axhline(y=0, linestyle="--")

plt.xlabel("Predicted House Value")
plt.ylabel("Residuals")
plt.title("Residual Plot")

plt.show()
```



7. Model Coefficients

The Linear Regression model calculates a coefficient for each input feature.

These coefficients indicate the direction and magnitude of the linear relationship between each feature and the predicted target, while the intercept represents the constant term of the regression equation.

The coefficients were calculated and displayed in the Jupyter Notebook.

8. Possible Improvements

The model can potentially be improved using the following approaches:

1. **Feature Engineering** — Create additional meaningful features from the available data.
2. **Feature Scaling** — Standardize numerical features where appropriate.

3. **Non-Linear Models** — Try models such as Random Forest or Gradient Boosting to capture non-linear relationships.
4. **Cross-Validation** — Use k-fold cross-validation for a more reliable performance estimate.
5. **Outlier Analysis** — Identify and investigate unusual observations that may affect model performance.
6. **Hyperparameter Tuning** — Tune parameters when using models that provide configurable hyperparameters.
7. **Additional Data** — Include additional relevant housing and geographic information.

9. Conclusion

In this project, a Linear Regression model was developed to predict California house values using the California Housing dataset.

The complete machine learning workflow was successfully implemented, including data exploration, preprocessing, train-test splitting, model training, prediction, evaluation, and visualization.

The model was evaluated using MAE, RMSE, and R^2 Score. The developed model provides a baseline for house value prediction.

Further improvements can be explored through feature engineering, cross-validation, outlier analysis, and non-linear machine learning algorithms.

10. Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook
- Linear Regression